Zewail City of Science and Technology

مدينة زويل للعلوم والتكنولوجيا

Sociology

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How Do We Do Sociology?

(Sociological Research Methods)



Introduction:

- Sociologists study societies on different levels:

1) **MICRO**: me (and my family) - Local soccer teams have a microculture.

2) **MESO**: national organizations, institutions, and ethnic subcultures.

3) **MACRO**:

Society: A nation has a national culture.

Global Community: Multinational organizations like the World Health Organization have a global culture.

- Researchers must always make clear how they collect their data. Every research study should be replicable— capable of being repeated—by other researchers. So, enough information must be given to ensure that another researcher could repeat the study and compare results

- The appropriate data collection method depends on the level(s) of analysis of the research question (micro, meso, or macro) the researcher is asking.
- For example, if you want to answer a macro-level research question, such as the effect of poverty on students dropping out of school in Egypt, you should focus on large-scale social and economic data sources such as the Egyptian census.
- To learn about micro-level issues such as the influence of peers on an individual's decision to drop out of school, you will need to examine small-group interactions at the micro level.



- Designing the Research Plan is vitally important because the researcher evaluates the various **methods** used to collect data for research studies and selects one or more that are appropriate for the research question.
- These include:

questionnaires,

interviews,

observational studies,

secondary data analysis,

content analysis,

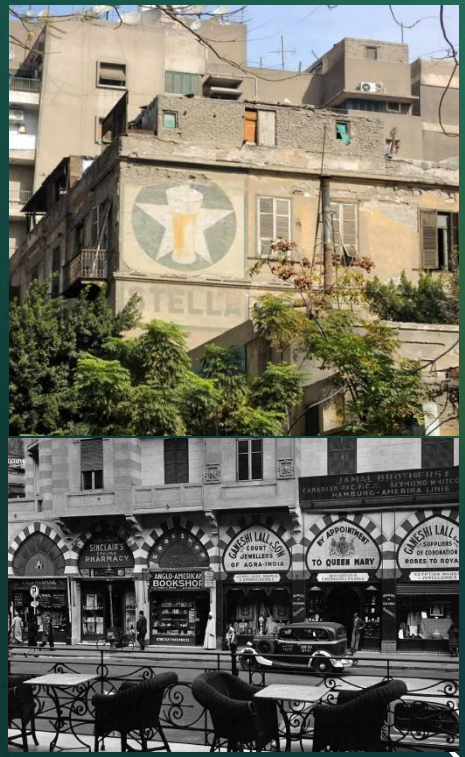
experiments.

- Some methods produce **quantitative (numerical) data** whereas others supply **qualitative (nonnumerical) data** such as individuals' responses to interviews.
- Questionnaires and secondary data analysis tend to be *quantitative* and used when conducting *macro-* and *meso-level* studies.
- Interviews, observational studies, and content analysis usually produce *qualitative* data or a *blend of quantitative and qualitative data* and are primarily used for *micro-level* research.



Being Clear About Causality

- Sociology as a science tries to be very careful about language—more precise than we usually are in our everyday conversations.
- What do we really mean when we say that something causes something else?



- At the heart of the research process is the effort to find **causal relationships** (i.e., one variable causes another one to change).

- The following key terms are important in understanding how two variables (concepts that vary in frequency and can be measured) are related:

CORRELATION

- **Correlation** refers to a relationship between variables (such as poverty and low levels of education), with change in one variable associated with change in another.
- The hypothesis mentioned earlier predicts that poverty and teenagers dropping out of school are related and vary together. That is, when the poverty level is high, dropping out of school is also high. If we claim that there is a correlation, however, that is only the first step. We have not yet established that change in one variable *causes* a change in the other.

CAUSE AND EFFECT VARIABLES

- Cause-and-effect relationships occur when there is a relationship between variables so that one variable stimulates a change in another.
- Once we have determined that there is probably a relationship, or correlation (the fact that the two variables, such as poverty and dropping out of school, both occur in the same situation), we need to take the next step: analyzing which comes first and seeing if one variable causes change in another.

- The independent variable is the variable in a cause-and-effect relationship that comes **first** in a time sequence and causes a change in another variable—the dependent variable.
- If we hypothesize that poverty causes Samy and others to drop out of school, poverty is the independent variable in this hypothesis and dropping out of school is the dependent variable, dependent on the poverty.

- In determining cause and effect, *the independent variable must always precede the dependent variable* in time sequence if we want to try to determine whether the independent variable *causes* a change in the dependent variable

SPURIOUS RELATIONSHIPS

- Spurious relationships occur when there is **no** causal relationship between the independent and dependent variables, but they vary together, often due to a third variable affecting both of them.

SPURIOUS RELATIONSHIPS

- For example, if the quantity of ice cream consumed is highest during those weeks of the year when most drownings occur, these two events are correlated. However, eating ice cream did not cause the increase in deaths.
- Indeed, hot weather may have caused more people both to purchase ice cream and to go swimming, with the larger number of swimmers resulting in more drowning incidents. The connection between ice cream and drownings is a spurious relationship.

Methods for Collecting Data:

1. Interviews are research conducted by talking directly with people and asking questions in person or by telephone.

2. Questionnaires contain questions and other types of items designed to solicit information appropriate to analysis of research questions (Babbie 2014). They are convenient for collecting large amounts of data because they can be distributed by mail or sent by e-mail to many respondents at once.

3. Observational studies (also called field research) involve systematic, planned observation and recording of interactions and other human behavior in their natural settings (where the activity normally takes place rather than in a laboratory).

They can take different forms:

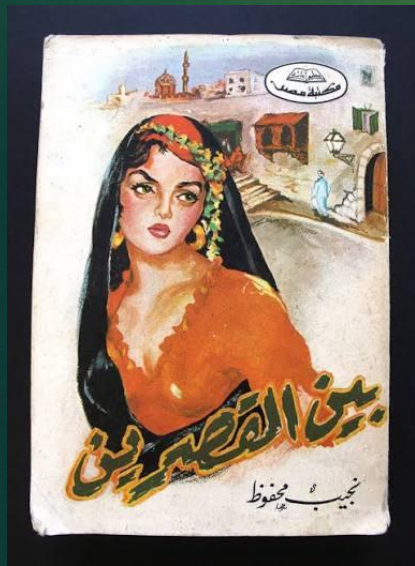
(1) observations in which the researcher actually participates in the activities of the group being studied or

(2) observations in which the researcher is not involved in group activities but observes or videotapes the activity. It is important for observers to avoid influencing or altering group functioning and interaction by their presence.

4. Secondary analysis uses existing data, information that has already been collected in other studies—including data banks such as the national census.

5. Content analysis entails the systematic categorizing and recording of information from written or recorded sources—printed materials, videos, radio broadcasts, or artworks.

- With content analysis (a common method in historical research), sociologists can gather the data they need from printed materials—books, magazines, newspapers, laws, letters, comments on websites, e-mails, videos, archived radio broadcasts, or even artwork.
- Content analysis of some novels may be very useful in revealing the culture and dynamics of the society.



6. In Experiments, all variables except the one being studied are controlled so researchers can study the effects on the variable under study.

- An experiment usually requires an experimental group, in which subjects in the group are exposed to the variable being studied to test the effects of that variable on human behavior, and a control group, in which the subjects are not exposed to the variable the researcher wants to test.

- The control group provides a baseline with which the experimental group can be compared.

- Control groups and experimental research projects outside of a lab setting are more common among sociologists.

- For example, researchers may want to determine whether a new teaching method using technology might help children from Samy's school.
- Researchers can do so by comparing a control group, exposed to the usual teaching, and an experimental group, provided with the new method or experimental technology.
- We must ensure that the control and experimental groups of children are at the same academic level and that the teachers are equally motivated and prepared when teaching both classes.

- With this carefully designed research project, we can conclude that the new approach increases learning if the children in the experimental group score significantly higher on the final exam than those in the control group.

Note that:

- The causes of human behavior are so complex that we frequently cannot do much more than form intelligent guesses about why or when certain behavior will occur.

- In many cases, individuals modify an aspect of their behavior in response to their awareness of being observed.

Ethical Issues in Social Research

Sociologists and other scientists are bound by the ethical codes of conduct governing research. The American Sociological Association (ASA) code of ethics outlines standards that researchers are expected to observe when doing research, teaching, and publishing. They include:

1. explaining the uses and consequences of the research and gaining informed consent from respondents;
2. taking steps to ensure the privacy of respondents;
3. being objective, reporting findings and sources fully;
4. making no promises to respondents that cannot be honored;
5. accepting no support that requires violation of these principles; and
6. delineating responsibilities in works with multiple authors.

Examples of unethical research include

- Studying people without their knowledge or consent,
- Only including data that support the results you would like to see, and
- Violating the confidentiality of your subjects by revealing their identity.

Thanks!